

SQL Indexes Explained

SQL Indexes

- You can create SQL indexes from single or multiple columns.
- A SQL index is like the index of a book. It speeds up retrieval of a record. The relational database management system (RDBMS) can retrieve a record with the index key instead of having to perform a *table scan*.
- MySQL automatically creates indexes for primary and foreign keys significantly speeding up join performance.
- You should only create indexes on columns used in a join or search because the RDMS must update index every time you execute an INSERT, UPDATE, or DELETE.

When to Create an Index

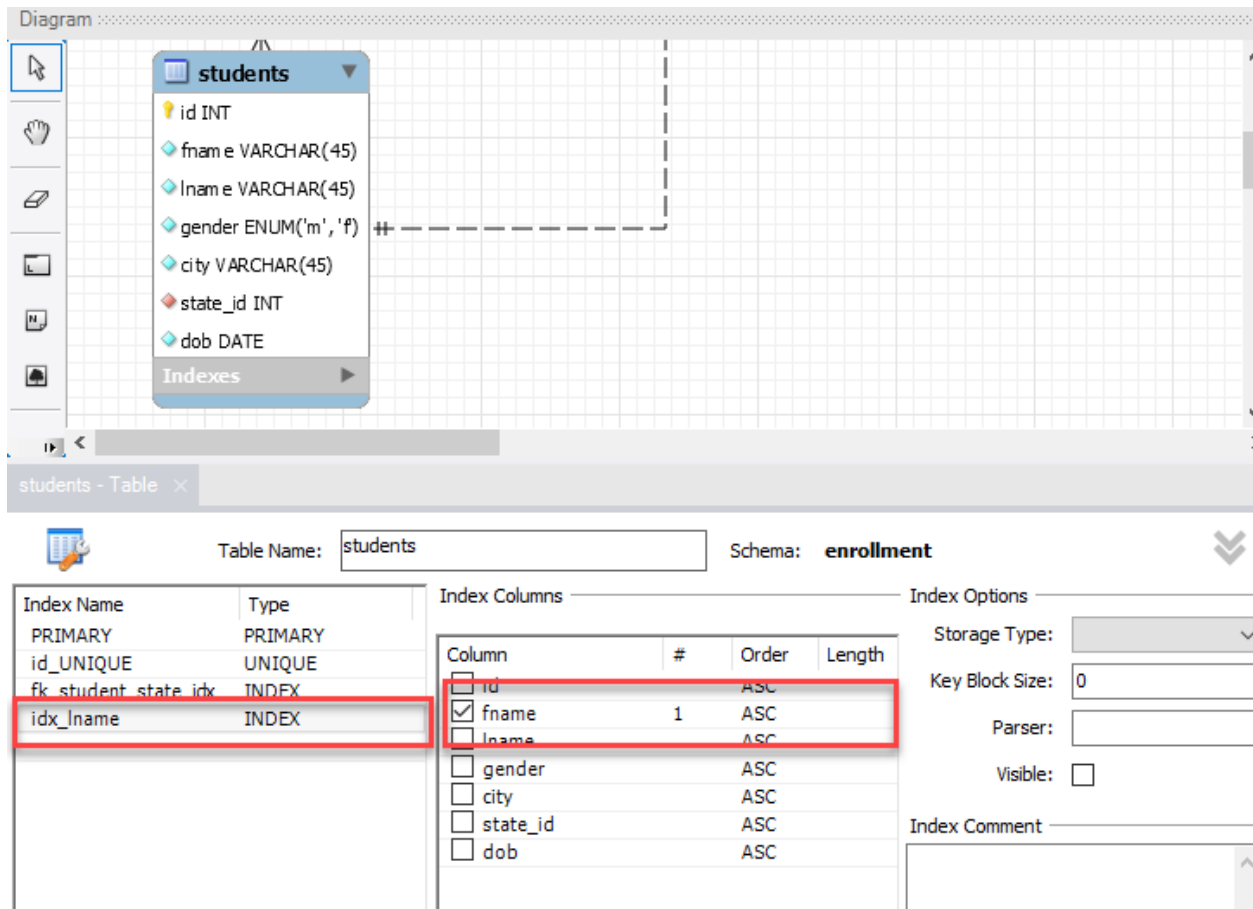
- When a column is used frequently in a search or a join.
- When a column contains a large number of distinct values.
- When the column is updated infrequently

Clustered vs Non-clustered indexes

- Clustered index: The values in the column indexed are physically stored in alphabetical or numeric order.
 - You can only have one clustered index per table.
 - If you assign a primary key, the system automatically creates a clustered index on that column.
 - If no primary key is defined on a table, the first column that has an index defined for it becomes the clustered index.
- Non-clustered index: Column values are not in alphabetical or numeric order
 - You can add as many non-clustered indexes to a table as you want.
 - You should only create additional non-clustered indexes on a table if you need to search or perform a join on that column. When you create a foreign key column, a non-clustered index is automatically created for that column.

How to Manually Create an Index in an ERD

- Right click on the table and select 'Edit'
- Click on the 'Indexes' tab
- Type the name of the index in the 'Index Name' field
- Under 'Type' select 'INDEX'
- Click on the column(s) that you want to index.
- Tab to a new line



Diagram

students

- id INT
- fname VARCHAR(45)
- lname VARCHAR(45)
- gender ENUM('m', 'f')
- city VARCHAR(45)
- state_id INT
- dob DATE

Indexes

students - Table

Table Name: **students** Schema: **enrollment**

Index Name	Type
PRIMARY	PRIMARY
id_UNIQUE	UNIQUE
fk_student_state_idx	INDEX
idx_lname	INDEX

Column	#	Order	Length
☐ id		ASC	
☒ fname	1	ASC	
☒ lname		ASC	
☐ gender		ASC	
☐ city		ASC	
☐ state_id		ASC	
☐ dob		ASC	

Index Options

Storage Type: **PRIMARY**

Key Block Size: **0**

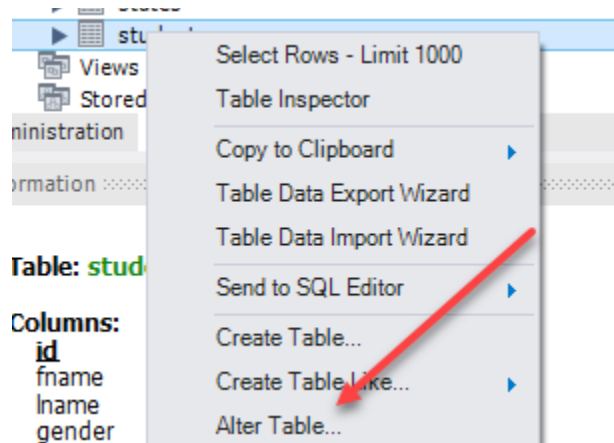
Parser: **Standard**

Visible: ☒

Index Comment

How to Manually Add an Index to an Existing Table

- Right click on the table
- Select 'Alter Table'



- Click on the 'Indexes' tab

Table Name: Schema: **enrollment**

Charset/Collation: Engine:

Comments:

Columns **Indexes** Foreign Keys Triggers Partitioning Options

- Type the name of the index in the 'Index Name' field
- Under 'Type' select 'INDEX'
- Click on the column(s) that you want to index.
- Tab to a new line

Table Name: Schema: **enrollment**

Charset/Collation: Engine:

Comments:

Index Name	Type
PRIMARY	PRIMARY
id_UNIQUE	UNIQUE
fk_student_state_idx	INDEX
idx_fname	INDEX

Index Columns

Column	#	Order	Length
☐ id		ASC	
☒ fname	1	ASC	
☐ lname		ASC	